MIL-M-62515(AT)
22 October 1985
SUPERSEDING
(see 6.3)

MILITARY SPECIFICATION

MAINTENANCE REPAIR KIT, ELECTRICAL, FOR M901A1 AND M981 WEAPON SYSTEMS

This specification is approved for use by U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a maintenance repair kit, herein referred to as the kit, used for the maintenance and repair of electrical connectors, contacts, and connector and cable accessories used on the weapon stations of the M901A1 and M981 weapon systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

MMM-A-121	- Adhesive, Bonding Vulcanized Synthetic Rubber to Steel.
MMM-A-130	- Adhesive, Contact.
MMM-A-132	- Adhesives, Heat Resistant, Airframe Structural, Metal-to-Metal.
MMM-A-134	- Adhesive, Epoxy Resin, Metal to Metal Structural Bonding.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.

MILITARY

MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-F-14072	– Finishes for Ground Electronic Equipment.
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive Back, General Specification for.
MIL-C-22520/2	- Crimping Tools, Terminal, Hand, Wire Termination for Wire Barrel Sizes 20 through 28.
MIL-C-22520/3	- Crimping Tools, Terminal, Hand, Wire Terminations; In-Service Inspection Gages.
MIL-C-22520/5	- Crimping Tools, Terminal, Wire Termination, Large for Coxial, Shielded Contacts and Ferrules, Terminal Lugs, Splices and End Caps,
MIL-I-23053/6	- Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Semi-Rigid, Crosslinked.
MIL-I-81969/8	- Installing and Removal Tools, Connector Electrical Contact, Types I and II, Class 2, Composition A.
MIL-I-81969/14	- Installing and Removal Tools, Connector Electrical Contact, Type III, Class 2, Composition B.
MIL-I-81969/17	- Installing and Removal Tools, Connector Electrical Contact, Type I, Class 1, Composition C.
MIL-I-81969/19	- Installing and Removal Tools, Connector Electrical Contact, Type II, Class 1, Composition C.
MIL-I-81969/30	- Installing and Removal Tools, Connector Electrical Contact, Type II, Class 2, Composition C.

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STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

12311946	- Maintenance Repair Kit.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish kits which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent kits delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in applicable specifications, standards, and drawings. When a definite material is not specified, the material used shall be suitable for the purpose and shall enable the kit to meet the performance requirements of this specification (see 4.8.1).

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3.2.1 Case. The case shall be fiberglass with an aluminum alloy frame (see 4.8.1).

3.2.2 Hinged covers and attaching components. The hinged covers and attaching components shall be aluminum alloy, clear anodized in accordance with MIL-F-14072 (see 4.8.1).

3.2.3 Clasps, handles, and hinges. The clasps, handles, and hinges shall be suitably protected to resist corrosion (see 4.8.1).

3.2.4 Inserts and retainer pads. The inserts and retainer pads shall be fire-retardant foamed polyethylene in accordance with PPP-C-1752 (see 4.8.1).

3.2.5 Instruction handbook sheets and contents charts. The instruction handbook sheets and contents charts shall be 0.020-inch thick white vinyl (see 4.8.1).

3.2.6 Adhesives. Adhesives shall be in accordance with MMM-A-121, MMM-A-130, MMM-A-132, or MMM-A-134 (see 4.8.1).

3.2.7 Dissimilar metals. The use of dissimilar metals in contact with each other shall be avoided unless suitably protected against electrolytic corrosion in accordance with requirement 16 of MIL-STD-454. Dissimilar metals are as specified in MIL–STD-889 (see 4.8.1).

3.3 Design and construction. The kit shall be designed and constructed as specified herein (see 4.5.2 and 4.8.2).

3.3.1 Configuration. The kit shall conform to figures 1 and 2 (see 4.5.2 and 4.8.2).

3.3.2 Color. The color of the case shall be olive drab in accordance with number 24087 of FED–STD-595 (see 4.5.2 and 4.8.2).

3.3.3 Rubber feet. Rubber feet shall be provided on the back and bottom of the case as shown in figures 1 and 2 (see 4.8.3).

3.3.4 Hinged covers. The hinged covers shall be as shown in figure 2. Each hinged cover shall have latch and a retainer pad (see figure 5) permanently affixed to the underside that will retain the tools in their respective positions when the hinged cover is closed. Each hinged cover shall have a warning label stating: “CAUTION: Lock this lid before closing” (see 4.8.3).

3.3.5 Instruction handbook and contents chart. The lettering shall be black, legible, and protected with a transparent coating. The instruction handbook shall contain a cover sheet with appropriate identification, detailed tool usage instructions and diagrams, and a connector part number breakdown appendix giving the shell sizes, insert arrangements, and the sizes and quantities of contacts found in the connectors serviced by the tool kit. The contents charts shall be centered on the inside of the top and bottom hinged covers, shall have a pressure-sensitive adhesive backing, and shall show the insert cavity configurations (see figures 3 and 4) and the respective tool part numbers (see 4.8.3).

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3.3.6 Foamed inserts. The foamed inserts shall be as shown in figures 6 and 7 (see 4.8.3).

3.3.7 Contents. The kit shall contain the items or their approved equivalents specified in table I and shown in figures 8 through 11, arranged as shown in figures 3 and 4 (see 4.8.3).

3.3.7.1 Tools. The tools shall conform to the corresponding specifications as shown in table I (see 4.8.1).

3.4 Environmental performance.

3.4.1 Low pressure (altitude). The kit with contents shall not be damaged when exposed to a pressure equivalent to an altitude of 40 000 feet (see 4.8.4.1).

3.4.2 High temperature. The kit with contents shall not be damaged when exposed to a temperature of 160 degrees Fahrenheit (°F) (see 4.8.4.2).

3.4.3 Low temperature. The kit with contents shall not be damaged when exposed to a temperature of minus 70°F (see 4.8.4.3).

3.4.4 Solar radiation (sunshine). The kit with contents shall not be damaged when exposed to solar radiation (see 4.8.4.4).

3.4.5 Rain. The kit with contents shall not be damaged, and there shall be no evidence of water or moisture accumulation inside the case, when exposed to blowing rain (see 4.8.4.5).

3.4.6 Humidity. The kit with contents shall not be damaged by exposure to 95 percent (%) relative humidity (see 4.8.4.6).

3.4.7 Fungus. The kit with contents shall not support fungus growth when exposed to conditions favorable for their development (see 4.8.4.7).

3.4.8 Salt fog. The kit with contents shall show no evidence of corrosion that could impair the operation of the kit and contents when exposed to a salt fog atmosphere (see 4.8.4.8).

3.4.9 Dust. The kit with contents shall not be damaged when exposed to blowing dust (see 4.8.4.9).

3.4.10 Leakage (immersion). The kit with contents shall not be damaged, and there shall be no evidence of water or moisture accumulation inside the case, when immersed in water to a depth of 3 feet (see 4.8.4.10).

3.4.11 Vibration. The kit with contents shall not be damaged when tested as specified in 4.8.4.11.

3.4.12 Shock. The kit with contents shall not be damaged when tested as specified in 4.8.4.12.

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TABLE I. Contents identification.

Item no. (location no.)	NSN	Part number	Manufacturer	Specification	Nomenclature	Qty
A1	5220-00-090-6722	M22520/3-1		MIL-C-22520/3	Inspection gage	1
A2	5120-00-165-3910	M22520/2-01		MIL-C-22520/2	Crimping tool frame	1
A3	5120-00-165-3913	M22520/2-02		MIL-C-22520/2	Positioner	1
A4	5120-00-017-3827	M22520-2-07		MIL-C-22520/2	Positioner	1
A5	5120-00-017-3927	M22520/2-09		MIL-C-22520/2	Positioner	1
A6	5120-00-017-3932	M22520/2-10		MIL-C-22520/2	Positioner	1
A7	5120-00-132-6913	M22520/5-01		MIL-C-22520/5	Crimping tool frame	1
A8		HX3-82	Daniels	See figure 8	Die removal tool	1
A9		Y554	Daniels		Die set	1
A10	5120-00-126-0860	M22520/5-100		MIL-C-22520/5	Die set	1
B1	5120-00-104-4066	MS3448-001		MIL-I-81969/30	Removal tool handle	1
B2	5120-00-148-9843	M81969/30-05		MIL-I-81969/30	Removal tool probe	1
B3		DRK105-(HDL)	Daniels	See figure 9	Removal tool handle	1
B4		DRK105-20-1	Daniels	See figure 9	Removal tool probe	1
B4		DRK105-22M-1	Daniels	See figure 9	Removal tool probe	1
B5	5120-00-915-4587	M81969/14-02; (MS3447-20)		MIL-I-81969/14	Installing/Removal tool	10
B6	5120-00-018-0535	M81969/14-02; (MS27534-20)		MIL-I-81969/14	Installing/Removal tool	10
B7	5120-00-018-0575	M81969/14-01; (MS27534-220)		MIL-I-81969/14	Installing/Removal tool	10

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TABLE I. Contents identification - Continued.

Item no. (location no.)	NSN	Part number	Manufacturer	Specification	Nomenclature	Qty
B8		DAK83-20	Daniels	See figure 10	Installing tool	1
B9		DRK83-20	Daniels	See figure 11	Removal tool	1
B10	5120-00-251-9503	M81969/8-01; (MS27495A22M)		MIL-I-81969/8	Installing tool	1
B11	5120-00-146-6558	M81969/8-02; (MS27495R22M)		MIL-I-81969/8	Removal tool	1
B12	5120-00-177-6967	M81969/8-05; (MS27495A20)		MIL-I-81969/8	Installing tool	1
B13	5120-00-177-6966	M81969/8-06; (MS27495R20)		MIL-I-81969/8	Removal tool	1
B14	5120-00-079-4598	M81969/17-09; (MS24256A20)		MIL-I-81969/17	Installing tool	1
B15	5120-00-079-4601	M81969/19-07; (MS24256R20)		MIL-I-81969/19	Removal tool	1
B16		DMC468-SCS-1	Daniels		Spare contact storage	1
B16		DMC468-SCS-2	Daniels		Spare contact storage	1
B17	5110-00-542-4487	744A			Wire stripper	1
B18		M23053/6		MIL-I-23053/6	Sleeving	--
--		DMC468-CS	Daniels	See 3.3.5	Instruction handbook	1
--		DMC218-01-5	Daniels	See figure 5	Retainer pad (attached to hinged covers)	2

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TABLE I. Contents identification - Continued.

Item no. (location no.)	NSN	Part number	Manufac-turer	Specification	Nomenclature	Qty
--		DMC468-1FM	Daniels	See figure 6	Insert (bottom)	1
--		DMC468-2FM	Daniels	See figure 7	Insert (top)	1
--		DMC468-CCB	Daniels	See 3.3.5	Contents chart (bottom)	1
--		DMC468-CCT	Daniels	See 3.3.5	Contents chart (top)	1
--		468672900-19	Daniels	See figures 1 and 2	Case	1

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3.5 Identification and marking. Identification and marking shall be in accordance with MIL-STD-130 and shall be permanent and legible (see 4.5.2 and 4.8.2).

3.5.1 Nameplates. The nameplates shall be marked with the information specified in figure 1, shall be 0.003- to 0.005-inch thick aluminum foil in accordance with MIL-P-19834, and shall be permanently mounted. The marking shall be black on a light background (see 4.5.2, 4.8.1, and 4.8.2).

3.6 Workmanship. Workmanship shall be in accordance with high-grade practices and quality and free of sharp edges, burrs, cracks, corrosion, or any other defect or feature which is a potential personal hazard to the operator during normal usage (see 4.5.2 and 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).
c. Control tests (see 4.6).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 + 18°F
b. Relative humidity 56 ± 30%
c. Barometric pressure 28.5 + 2.0 inches mercury (Hg)
- 3.0

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select one kit produced under the production contract for first article inspection. First article samples shall be inspected as specified in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply kits that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor, Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Materials and construction	3.2 thru 3.2.7, 3.3.7.1 and 3.5.1	4.8.1	X			
Defects (see 4.5.2 and table III)	3.3 thru 3.3.2 and 3.5 thru 3.6	4.8.2	X	X		
Features	3.3.3 thru 3.3.7	4.8.3	X		X	
Low pressure (altitude)	3.4.1	4.8.4.1	X			
High temperature	3.4.2	4.8.4.2	X			
Low temperature	3.4.3	4.8.4.3	X			

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TABLE II. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Solar radiation (sunshine)	3.4.4	4.8.4.4	X			
Rain	3.4.5	4.8.4.5	X			
Humidity	3.4.6	4.8.4.6	X			X
Fungus	3.4.7	4.8.4.7	X			
Salt fog	3.4.8	4.8.4.8	X			X
Dust	3.4.9	4.8.4.9	X			
Leakage (immersion)	3.4.10	4.8.4.10	X			X
Vibration	3.4.11	4.8.4.11	X			
Shock	3.4.12	4.8.4.12	X			X

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the kits of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

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TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
103	Identification marking, missing, incomplete, or not legible (see 3.5 and 3.5.1).	Visual
104	Faulty workmanship affecting performance (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
202	Color, improper (see 3.3.2).	Visual
203	Identification marking, defective, improper location or size (see 3.5 and 3.5.1).	Visual
204	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Test. Each kit shall be subjected to the quality conformance tests specified in table II.

4.6 Control tests. Control tests shall be conducted on one kit from each lot of 50 units consecutively produced, except that not more than one test shall be performed in a 3-month period, nor less than one test in a 6-month period. The kit shall be subjected to the tests specified in table II.

4.7 Failure. Failure of any kit to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.2.7, 3.3.7.1, and 3.5.1 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.8.2 Defects. Conformance to 3.3 through 3.3.2 and 3.5 through 3.6 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Features. Conformance to 3.3.3 through 3.3.7 shall be determined by exercising the kit, and by qualitative observation sufficient to demonstrate that specified characteristics and features are present and functional.

4.8.4 Environmental performance tests.

4.8.4.1 Low pressure (altitude). To determine conformance to 3.4.1, test the kit with contents as specified in MIL-STD-81OD, method 500.2, procedure I. The pressure shall be equivalent to an altitude of 40 000 feet and the duration one hour. After the test, examine the kit and contents for damage.

4.8.4.2 High temperature. To determine conformance to 3.4.2, test the kit with contents as specified in MIL-STD-810D, method 501.2, procedure I. The temperature shall be 160°F and the duration four hours. After the test, examine the kit and contents for damage.

4.8.4.3 Low temperature. To determine conformance to 3.4.3, test the kit with contents as specified in MIL-STD-810D, method 502.2, procedure I. The temperature shall be minus 70°F and the duration 24 hours. After the test, examine the kit and contents for damage.

4.8.4.4 Solar radiation (sunshine). To determine conformance to 3.4.4, test the kit with contents as specified in MIL-STD-81OD, method 505.2, procedure I. The test shall be for three 24–hour cycles. After the test, examine the kit and contents for damage.

4.8.4.5 Rain. To determine conformance to 3.4.5, test the kit with contents as specified in MIL-STD-810D, method 506.2, procedure I. The rainfall rate shall be 4 inches per hour and the wind velocity 40 miles per hour. After the test, examine the kit and contents for damage and for any evidence of water or moisture accumulation inside the case.

4.8.4.6 Humidity. To determine conformance to 3.4.6, test the kit with contents as speciifed in MIL-STD-202F, method 103B. After the test, examine the kit and contents for damage.

4.8.4.7 Fungus. To determine conformance to 3.4.7, test the kit with contents as specified in MIL-STD-810D, method 508.3. The external surfaces shall be sprayed with spore suspension (as defined by the “specimen inoculation” paragraph of MIL-F-13927) followed by exposure to ambient air temperatures of 80 to 84°F at 96 to 100% relative humidity for 28 days. After the test, examine the kit and contents for any evidence of fungus growth.

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4.8.4.8 Salt fog. To determine conformance to 3.4.8, test the kit with contents as specified in MIL-STD-202F, method 101D, test condition letter B. After the test, examine the kit and contents for any evidence of corrosion that could impair its operation.

4.8.4.9 Dust. To determine conformance to 3.4.9, test the kit with contents as specified in MIL-STD-810D, method 510.2, procedure I. The air velocity shall be 1750 + 250 feet per minute with a dust concentration of 0.3 + 0.2 gram per cubic feet. The dust size shall be as follows:

100% shall pass through a 100 mesh screen.
98 ± 2% shall pass through a 140 mesh screen.
90 ± 2% shall pass through a 200 mesh screen.
75 ± 2% shall pass through a 325 mesh screen.

After the test, examine the kit and contents for damage.

4.8.4.10 Leakage (immersion). To determine conformance to 3.4.10, test the kit with contents as specified in MIL-STD-810D, method 512.2. The depth shall be 3 feet and the duration 1 hour. After the test, examine the kit and contents for damage and any accumulation of water or moisture inside the case.

4.8.4.11 Vibration. To determine conformance to 3.4.11, test the kit with contents as specified in MIL-STD-810D, method 514.3, category 3, procedure II. The test duration shall be 3 hours. After the test, examine the kit and contents for damage.

4.8.4.12 Shock. To determine conformance to 3.4.12, test the kit with contents as specified in MIL-STD-810D, method 516.3, procedure I. After the test, examine the kit and contents for damage.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The kit and contents therein covered by this specification are intended to be used for the repair and maintenance of the M901A1 and M981 weapon station electrical connections.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of applicable drawings.
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).

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f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article inspection is not required (see 4.4).
h. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Supersession data. This military specification supersedes M901A1PD-62396, dated 15 February 1984.

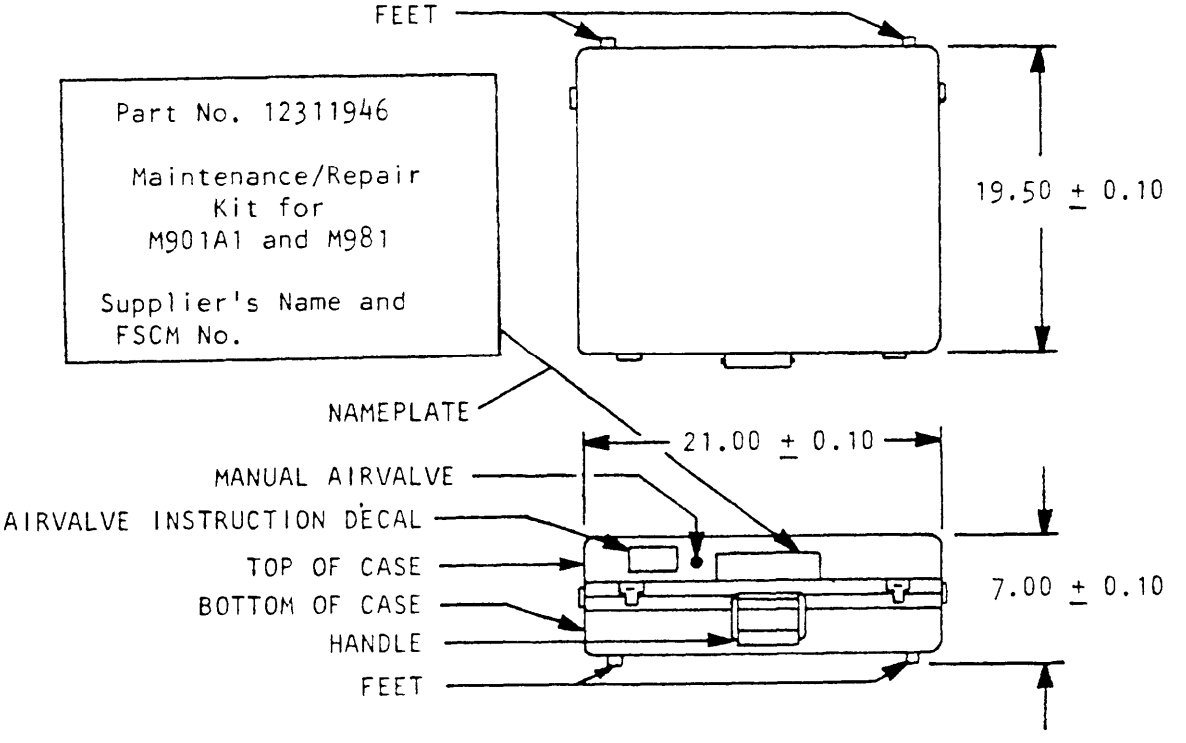
6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged.

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NOTES: 1. Dimensions are in inches.
2. Remove all burrs, sharp edges, and corners.

FIGURE 1. Maintenance repair kit for M901A1 and M981 weapon station connector systems.

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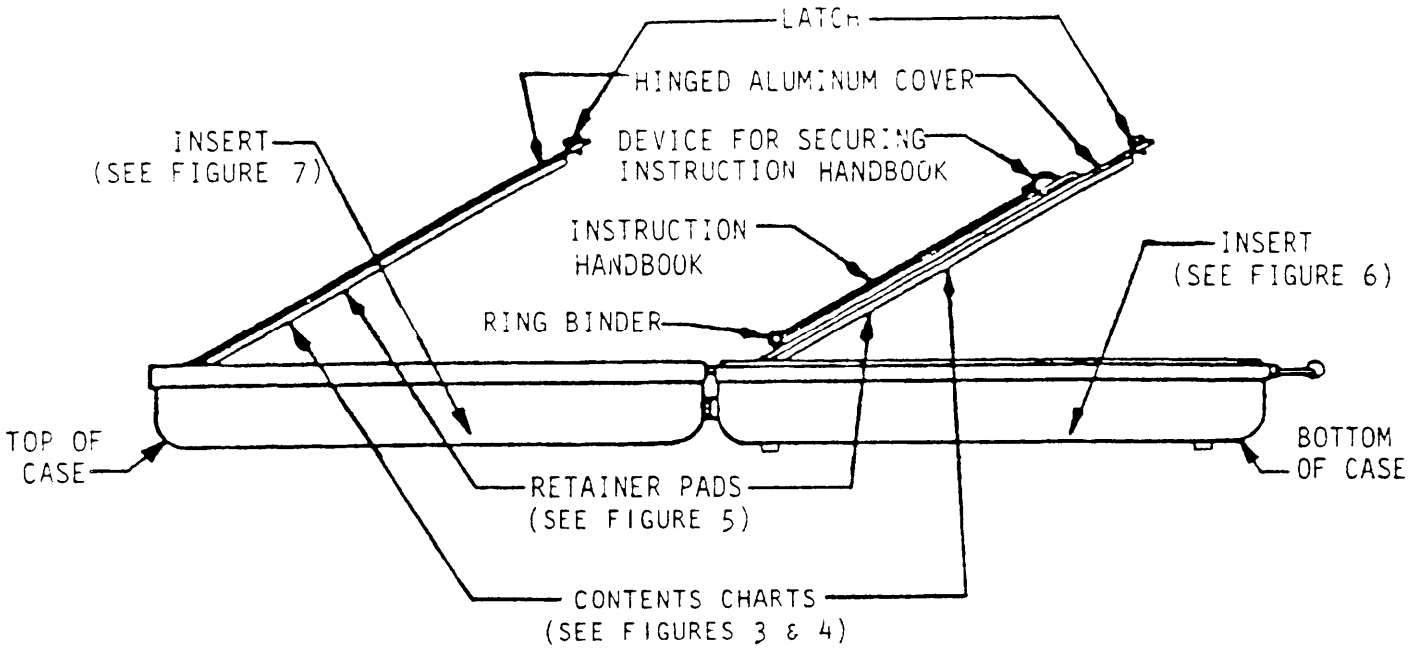
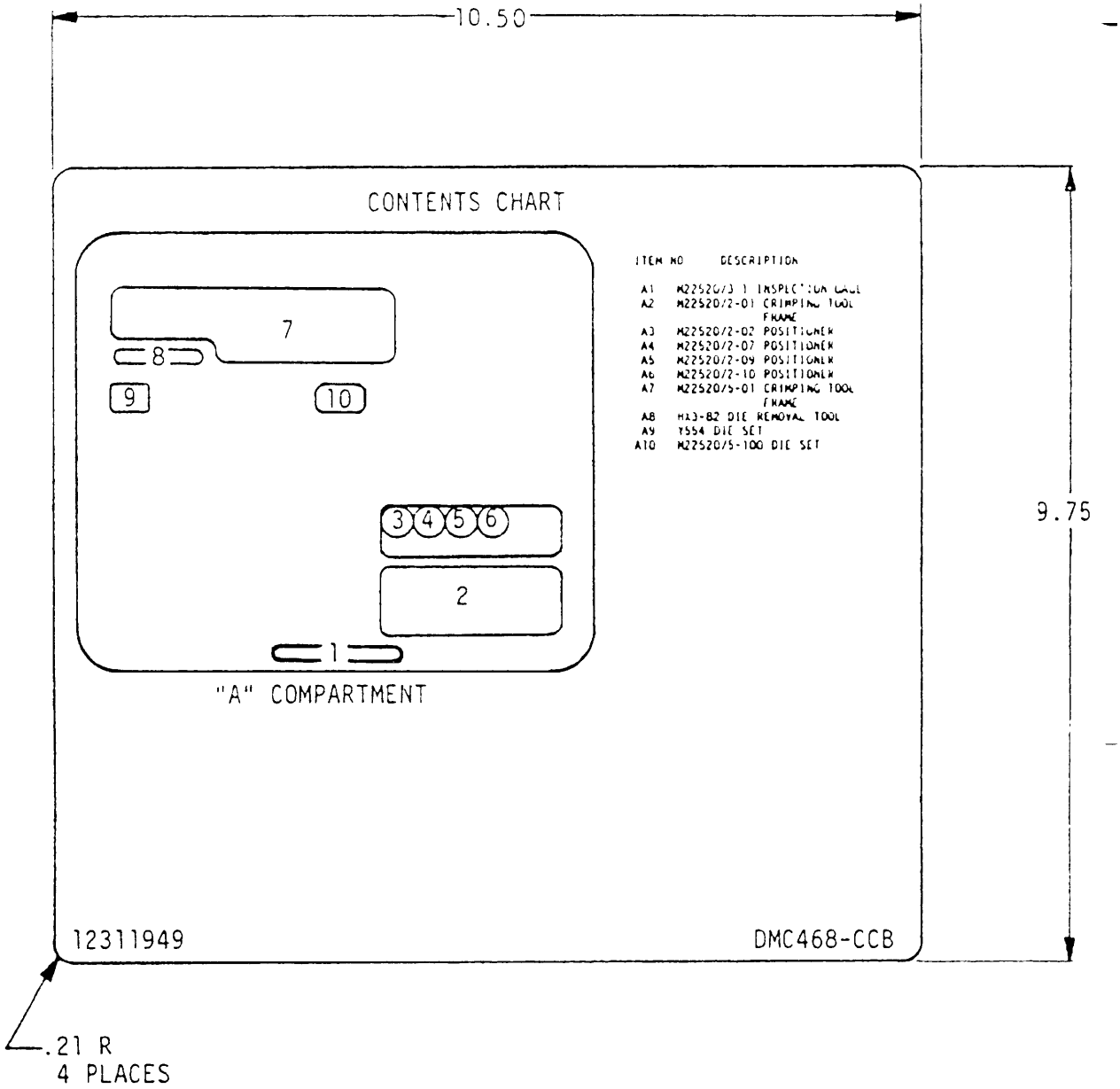


FIGURE 2. Maintenance repair kit assembly.

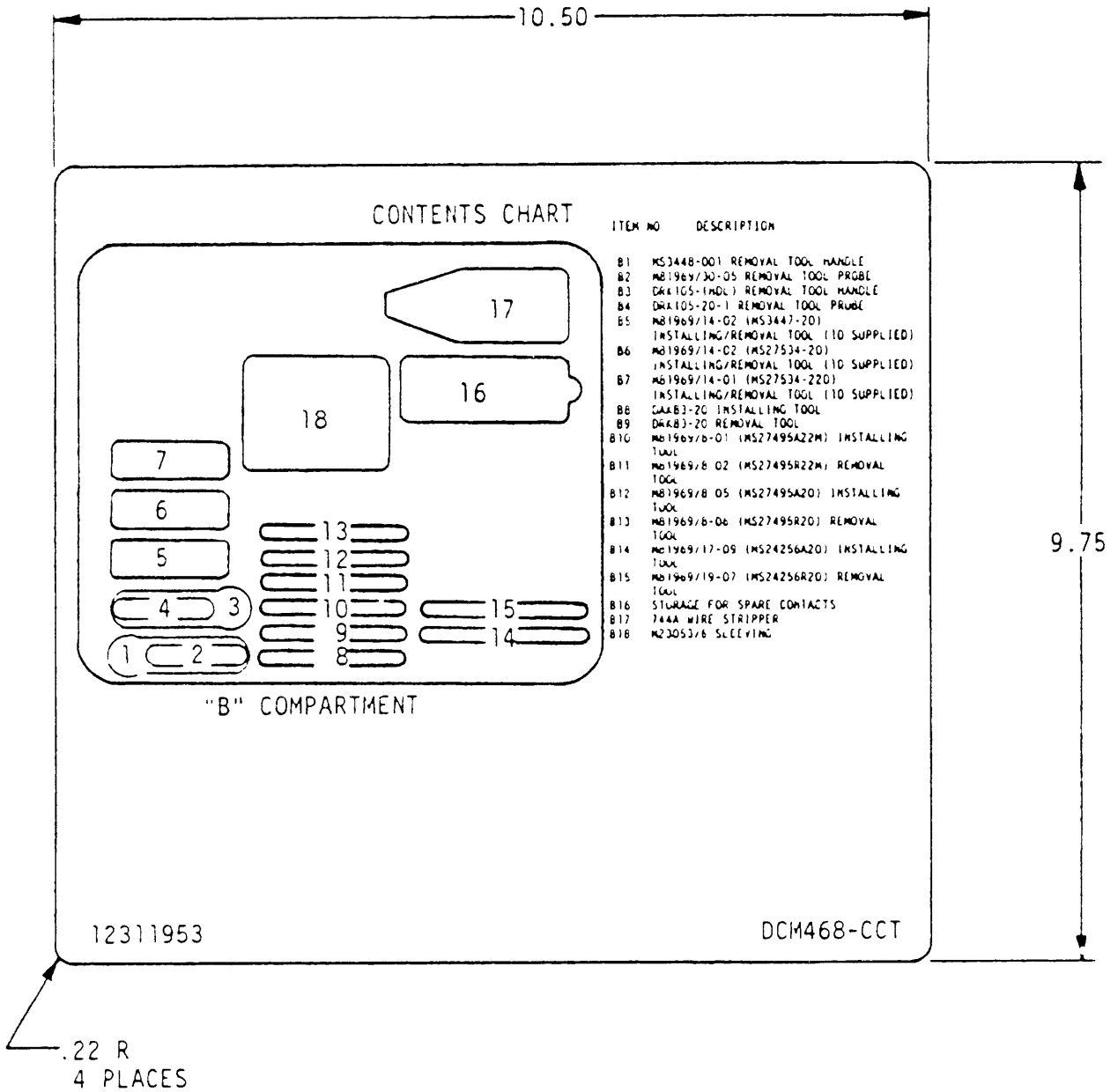
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NOTE: Dimensions are in inches.

FIGURE 3. Contents chart for bottom of case.

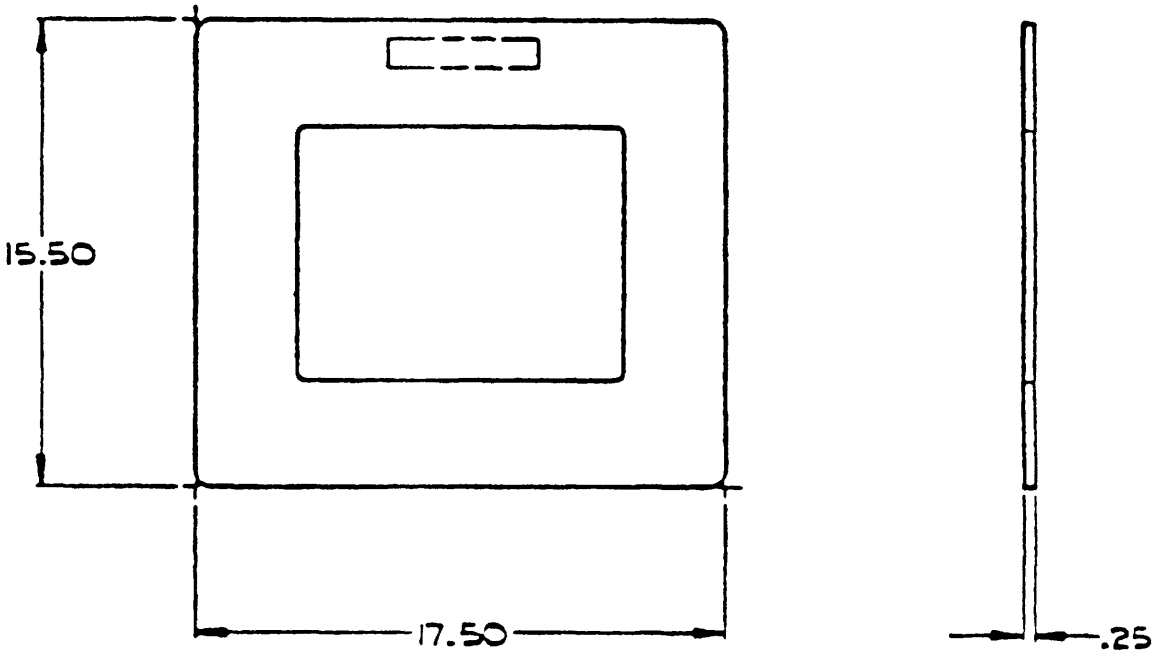
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NOTE: Dimensions are in inches.

FIGURE 4. Contents chart for bottom of case.

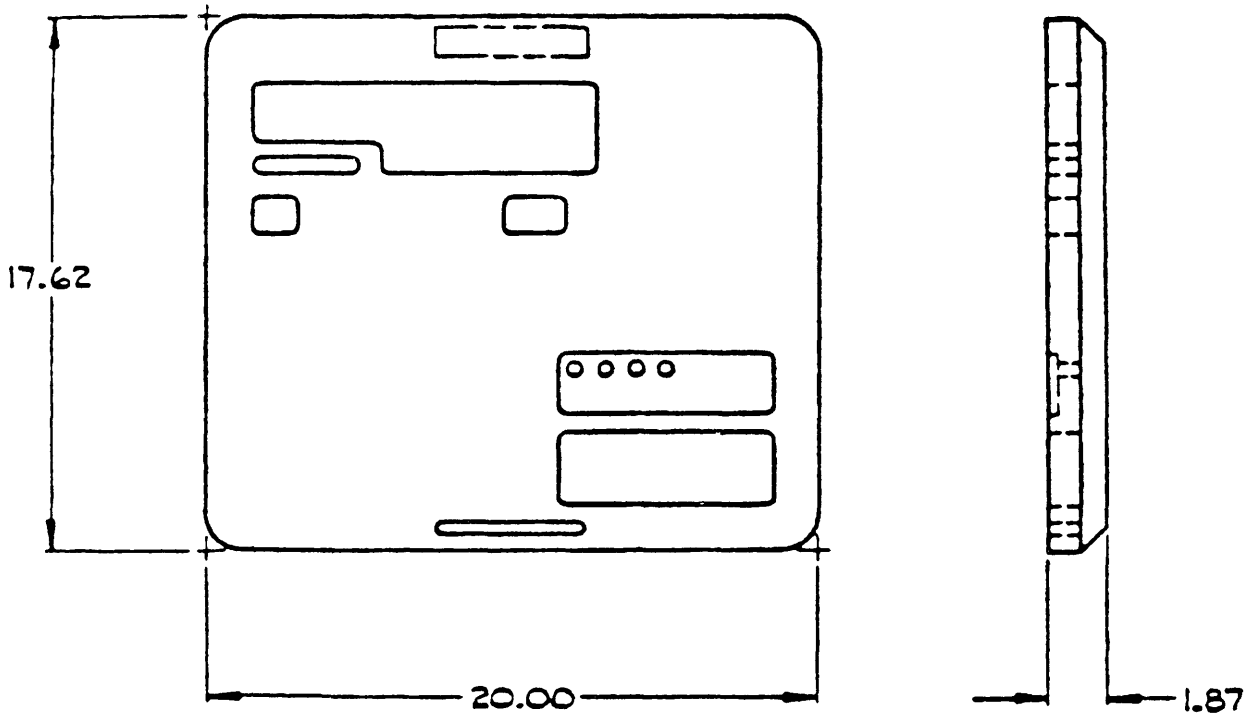
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NOTE: Dimensions are in inches.

FIGURE 5. Retainer pad.

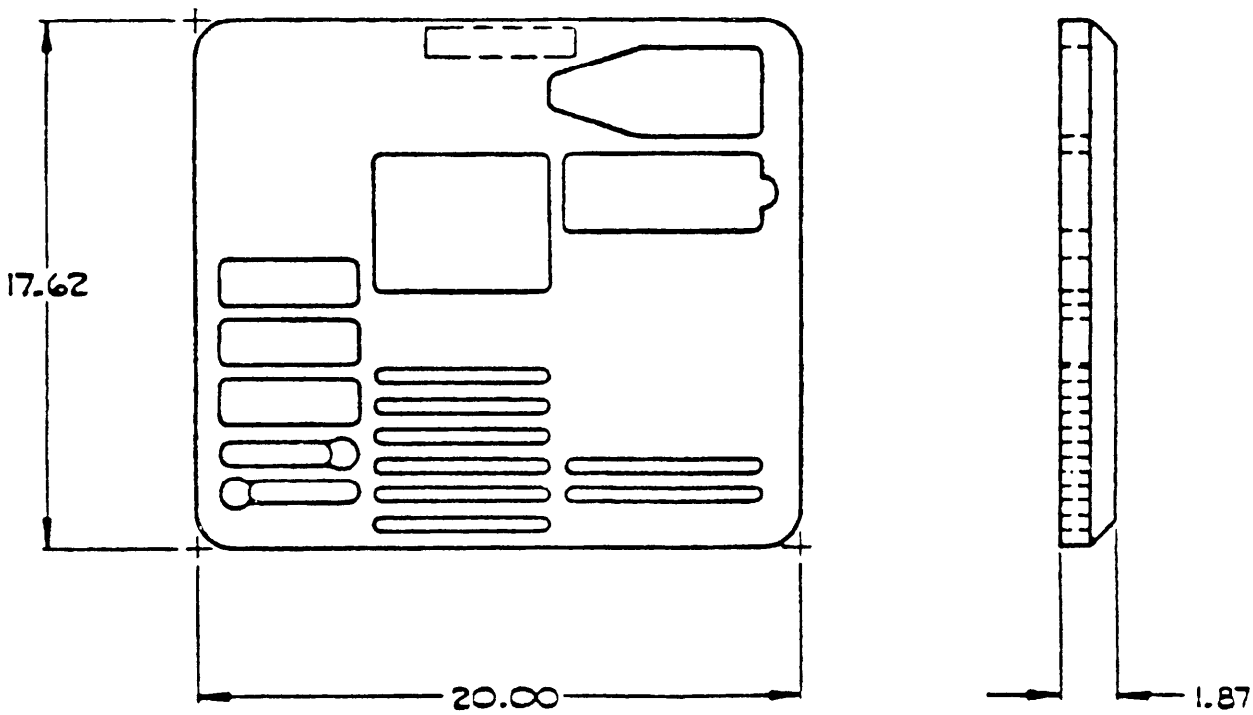
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NOTE: Dimensions are in inches.

FIGURE 6. Foamed insert for bottom of case.

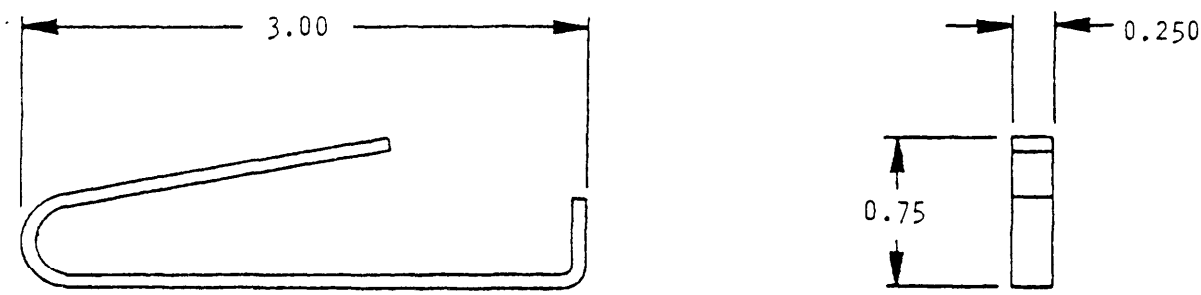
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NOTE: Dimensions are in inches.

FIGURE 7. Foamed insert for top of case.

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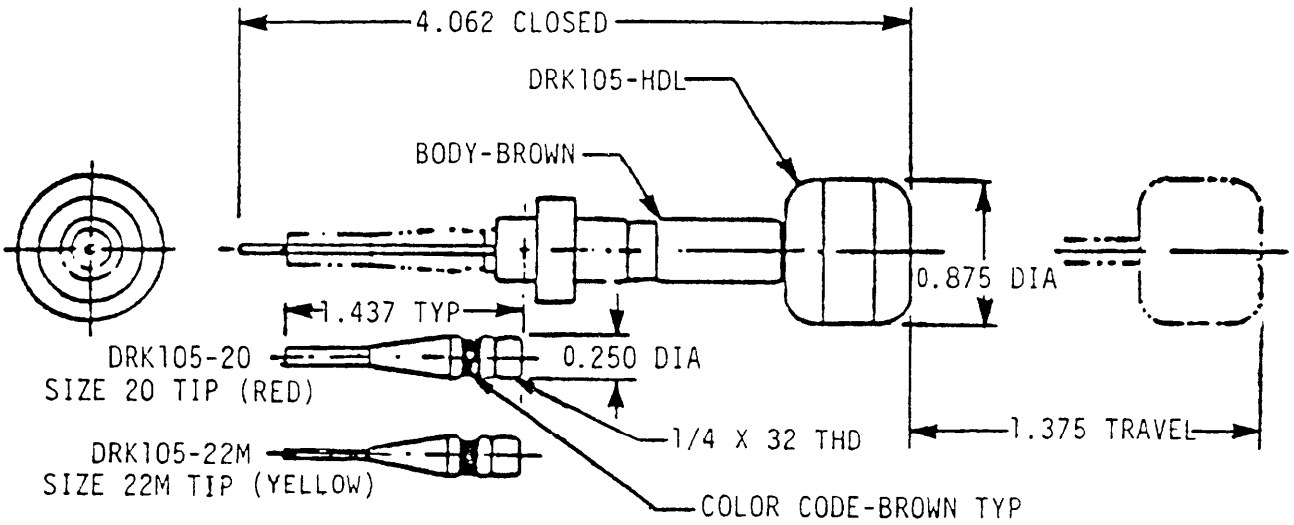
NOTES:

1. HX3-82 die knock-out tool is used to remove die sets from M22520/5-01 crimp tool.

2. Dimensions are in inches.

FIGURE 8. Die removal tool.

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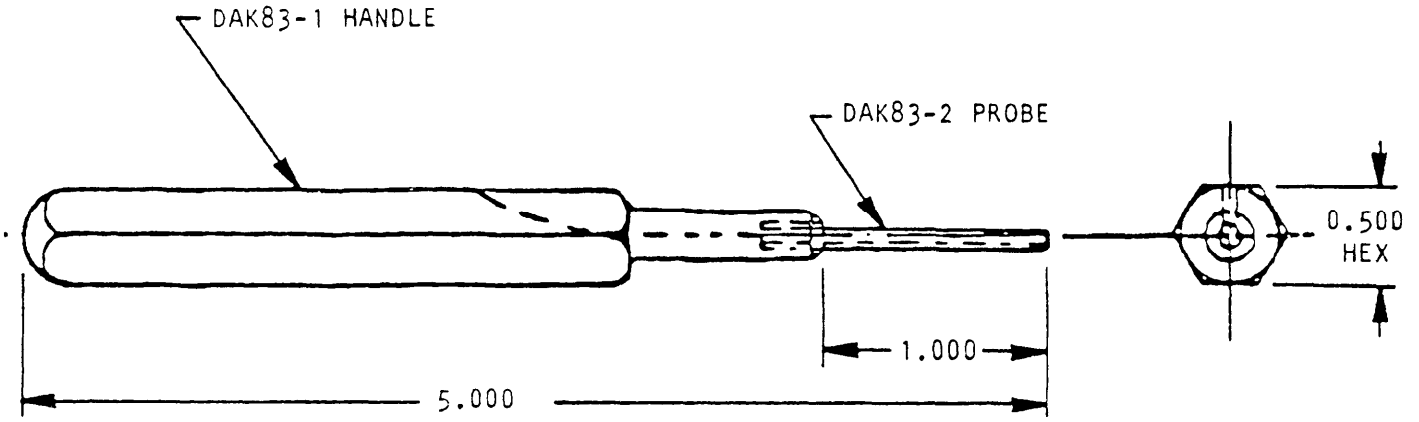
NOTES:

1. Daniels DRK105 removal tool is used for MIL-C-38999 unwired contacts.

2. Dimensions are in inches.

FIGURE 9. Removal tool set.

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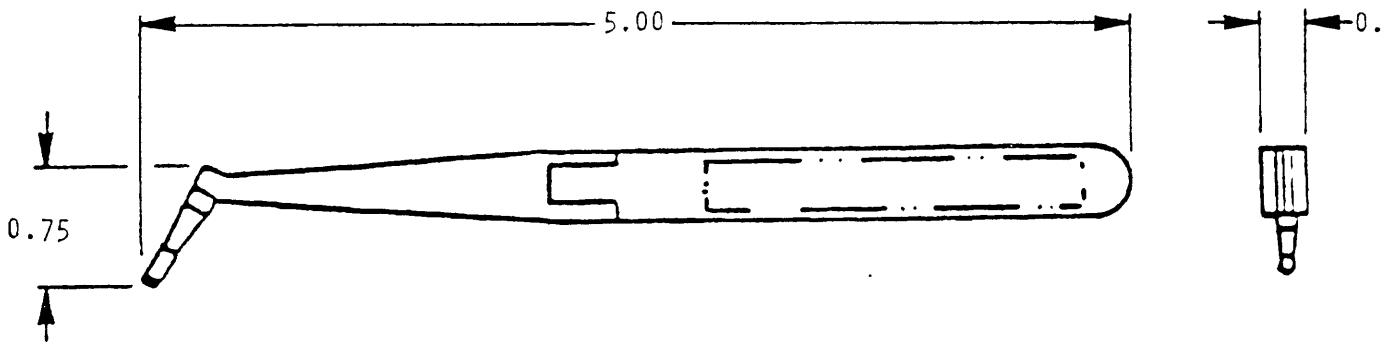
NOTES:

1. Daniels DAK83-20 installing tool is used to insert M39029/4-20-20 and M39029/5-20-20 contacts in MIL-C-26482 or MIL-C-83723 circular connectors.

2. Dimensions are in inches.

FIGURE 10. Installing tool.

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NOTES:

1. Daniels DRK83-20 removal tool is used to remove size 20 contacts from MIL-C-5015, MIL-C-26482, and MIL-C-83723 connectors.

2. Dimensions are in inches.

FIGURE 11. Removal tool.

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